

Deadlock : IDP Login – Performance Test Strategy & Deadlock Issue Tracking

Owner: QA / Performance / Auth-Server Teams
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Related Docs:

- [Deadlock Tiger Team Notes / Teams Chat](#)
- [RT Ticket #501630](#)
- [JMeter Test Planning Doc](#)

Overview

The IDP login flow began exhibiting performance degradation and **deadlock symptoms**, causing login-related timeouts and system instability. This impacted both Stage1 and Production environments intermittently. A Tiger Team was assembled to analyze the root cause, coordinate patches, and validate performance improvements.

Identified Issues

Date	Summary	Notes
Aug 25	Entity login redirect failure, blank pages post-MFA, 400 errors on tab duplication	Traced to session handling conflicts introduced by earlier fixes
Aug 26–27	DispatcherServlet overload	Static content (80+ calls per login page) was hitting DispatcherServlet, identified via New Relic
Aug 27	Code fix deployed to Stage1	Patch v1.13.6860 tested, improved stability confirmed
Aug 28–29	Patch validated in PROD	Functionalities verified: Enrollment, Login, Logout, Profile Update, Mobile Login (incl. Biometric)

Root Cause Summary

- **DispatcherServlet Overload:** Excessive requests for static content (CSS, JS, images) were not bundled/minified.
- **Session Management Conflicts:** Improper cleanup and handling of session state across tabs caused erratic behavior.
- **Deadlock Symptoms:** Code execution stuck under certain flows, traced to the auth-server and fixed via targeted patch.

Performance Strategy & Testing Approach

Objective	Strategy
Simulate realistic login behavior	Augment JMeter scripts with explicit static content fetch (CSS, JS)
Stress DispatcherServlet endpoint	Add focused GET calls in test scripts for large resource bundles
Validate PKCE flow (CS/API hybrid)	Ensure token exchange & login redirect are tested in correct sequence
Track impact via APM	Use New Relic for DispatcherServlet-specific monitoring
Performance Baseline	Compare Stage1 performance pre- and post-patch using BlazeMeter Dashboards
Browser behavior emulation	Enhance scripts to reflect browser-like multi-request login page loading

Fix Deployment Timeline

Date	Environment	Activity
Aug 27	Stage1	Patch v1.13.6860 deployed, final version tested
Aug 28 (9 PM)	PROD – Step 1	Auth-server patch deployed
Aug 29 (5 AM)	PROD – Step 2	uiapp patch reverted to fix redirect logic
Aug 30	QA Signoff	Confirmed login, logout, MFA, profile update, mobile login flows in all plans

Roles & Responsibilities

Name	Role	Responsibility
Abhitosh Fani	Dev Lead	Patch creation, testing, and deployment
Nick Botvinnik / Swapnil Patil	Perf Engg.	JMeter strategy, static content fetch testing
Pramod Gundelly	QA	Validation across plans and mobile
Rajib Akhter	Stakeholder	Performance testing coordination
Henry Dittmer / Mark Foong	Management	Risk tracking, prioritization, approvals

Next Steps

- Finalize **DispatcherServlet stress strategy**
- Add **JMeter static content calls** for full page simulation
- Monitor APM traces and compare **before/after metrics**
- Document long-term optimization proposals (e.g., resource bundling)
- Confirm cross-tab session management fix is stable
- Coordinate 4th patch if needed based on residual issues

Conclusion

The Tiger Team approach has successfully stabilized the IDP login flow through patch coordination, targeted testing, and collaborative debugging. Continued performance validation will ensure readiness for upcoming releases involving more plan migrations to IDP.